

Object Detection Algorithm for Autonomous Driving: Implementation and Performance Study of the AttenRetina Model Learning

*Report submitted to the SASTRA Deemed to be University
as the requirement for the course*

CSE300 - MINI PROJECT

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MAY 2026



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Bonafide Certificate

This is to certify that the report titled “**Object Detection Algorithm for Autonomous Driving: Implementation and Performance Study of the AttenRetina Model**” submitted as a requirement for the course, **CSE300 : MINI PROJECT** for B.Tech. is a bonafide record of the work done by **Ms. B. Sai Eshita Reddy M** (Reg. No.127003032, B. Tech Computer Science and Engineering), **Mr. Rahul R** (Reg. No.127003201, B. Tech Computer Science and Engineering) and **Ms. Rajalakshmi R** (Reg. No.127003202, B. Tech Computer Science and Engineering) during the academic year 2025-26, in the School of Computing, under my supervision.

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Date : 06/05/2026

Mini Project *Viva voice* held on: 08/05/2026

Examiner 1

Examiner 2

ACKNOWLEDGEMENTS

We would like to thank our Honorable Chancellor **Prof. R. Sethuraman** for providing us with an opportunity and the necessary infrastructure for carrying out this project as a part of our curriculum.

We would like to thank our Honorable Vice-Chancellor **Dr. S. Vaidhyasubramaniam** and **Dr. S. Swaminathan**, Dean, Planning & Development, for the encouragement and strategic support at every step of our college life.

We extend our sincere thanks to **Dr. R. Chandramouli**, Registrar, SASTRA Deemed to be University for providing the opportunity to pursue this project.

We extend our heartfelt thanks to **Dr. V. S. Shankar Sriram**, Dean, School of Computing, **Dr. R. Muthaiah**, Associate Dean, Research, **Dr. K. Ramkumar**, Associate Dean, Academics, **Dr. D. Manivannan**, Associate Dean, Infrastructure, **Dr. R. Alageswaran**, Associate Dean, Students Welfare

Our guide **Dr. Divya Lakshmi K**, Assistant Professor - III, School of Computing was the driving force behind this whole idea from the start. Her deep insight in the field and invaluable suggestions helped us in making progress throughout our project work. We also thank the project review panel members for their valuable comments and insights which made this project better.

We would like to extend our gratitude to all the teaching and non-teaching faculties of the School of Computing who have either directly or indirectly helped us in the completion of the project.

We gratefully acknowledge all the contributions and encouragement from my family and friends resulting in the successful completion of this project. We thank you all for providing us an opportunity to showcase our skills through this project.

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ABBREVIATIONS

ML – Machine Learning

DL – Deep Learning

CNN – Convolutional Neural Network

FPN – Feature Pyramid Network

mAP – Mean Average Precision

IoU – Intersection over Union

KITTI – Autonomous Driving Dataset

COCO – Common Objects in Context

NOTATIONS

- X – Input image dataset
- N – Number of training samples
- F – Feature map extracted from backbone network F_i
- Feature map at level i in FPN
- P_i – Pyramid feature at level i y_i
- Ground truth label
- $\hat{y}_i$ – Predicted label
- Lcls – Classification loss
- Lbox – Bounding box regression loss
- α – Balancing parameter in focal loss
- γ – Focusing parameter in focal loss
- IoU – Intersection over Union
- mAP – Mean Average Precision

Abstract

Autonomous driving systems rely on accurate object detection to identify surrounding elements such as vehicles, pedestrians, cyclists, and traffic signs. Although deep learning-based object detection techniques have made significant progress, real-world traffic environments remain challenging due to complex backgrounds, scale variations, occlusions, illumination changes, and the presence of small or distant objects. Conventional detection models, including Faster R-CNN, SSD, YOLO, and RetinaNet, perform well on large and clearly visible targets but often show reduced accuracy for small objects in complex scenes. Moreover, the integration of multi-scale feature fusion and attention mechanisms in recent models increases computational complexity, limiting their suitability for real-time and resource-constrained deployment. This paper presents the implementation and evaluation of an enhanced AttenRetina object detection model that addresses these challenges. Building on the original AttenRetina framework, the proposed approach incorporates a computationally efficient attention mechanism to reduce model complexity while retaining the advantages of multi-scale feature fusion through the Feature Pyramid Network (FPN). To improve robustness in challenging environments, the model is further evaluated under occlusion and low-light conditions. The enhanced AttenRetina model is evaluated on the KITTI and MS COCO datasets using standard metrics such as Precision, Recall, F1-score, and mean Average Precision (mAP). Experimental results demonstrate improved detection accuracy and robustness for small objects and adverse visual conditions, while maintaining reduced computational overhead, highlighting its suitability for autonomous driving applications.

Keywords— Autonomous Driving, Object Detection, AttenRetina, Feature Pyramid Network (FPN), Attention Mechanism, Deep Learning, Small Object Detection, Multi-Scale Feature Fusion, Real-Time Detection, KITTI Dataset, MS COCO Dataset, Low-Light Detection, Occlusion Handling, Mean Average Precision (mAP).

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CHAPTER 1

SUMMARY OF BASE PAPER

1.1 Overview

Autonomous driving systems rely heavily on accurate and real-time object detection to ensure safe navigation and decision-making. Detecting objects such as pedestrians, vehicles, and traffic signs in complex environments is a critical challenge. The AttenRetina model is proposed to improve object detection performance by combining multi-scale feature extraction and attention mechanisms.

1.2 Problem Statement

Existing object detection models face several challenges:

- Difficulty in detecting small and distant objects
- Performance degradation in complex environments
- Handling occlusion and lighting variations
- Trade-off between accuracy and real-time performance

These limitations reduce the reliability of autonomous driving systems.

1.3 Objectives

The objectives of this project are:

- To develop an efficient object detection model
- To improve detection of small and distant objects
- To enhance feature extraction using multi-scale learning
- To integrate attention mechanisms for improved accuracy
- To achieve a balance between accuracy and real-time performance

1.4 Proposed Solution

The proposed system uses the AttenRetina model, which combines:

- RetinaNet for object detection
- Feature Pyramid Network (FPN) for multi-scale feature extraction
- Attention mechanisms (spatial and channel attention)

This approach improves detection accuracy and robustness in complex environment.

1.5 Architecture

The system architecture consists of the following components :

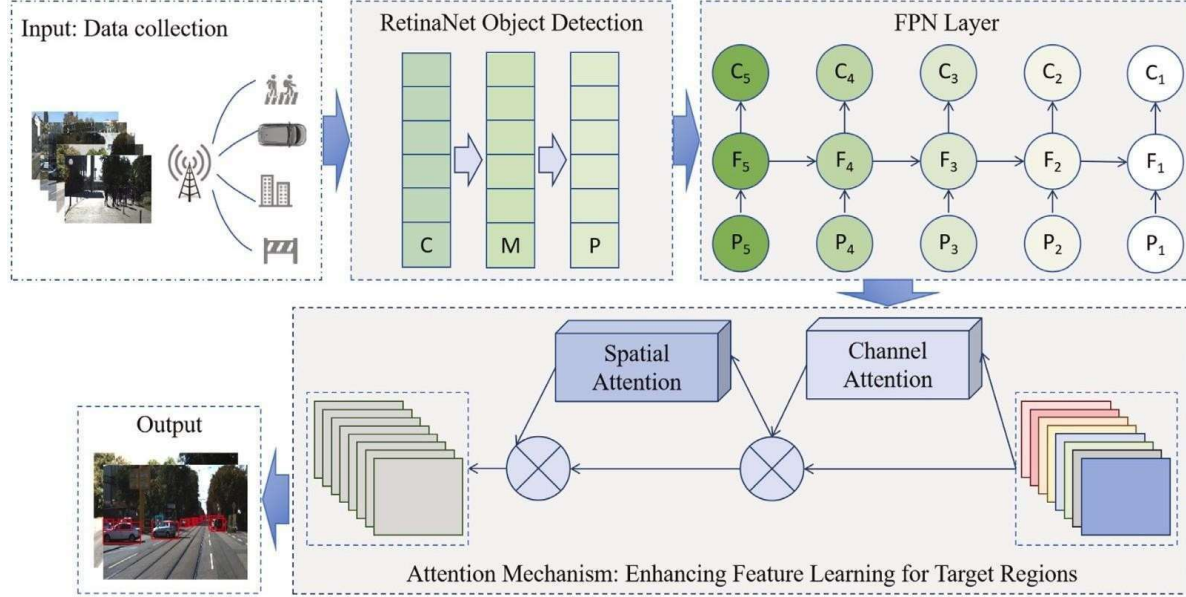


Fig 1.1: System Architecture

The above figure illustrates the complete workflow of the proposed model, including feature extraction, multi-scale feature fusion, and attention-based refinement for accurate object detection.

1. Input Image
2. Backbone Network (ResNet)
3. Feature Pyramid Network (FPN)
4. Attention Modules
5. Detection Output

The FPN enhances multi-scale feature learning, while attention modules help focus on important regions in the image. Input image is given to the **backbone (ResNet)** for feature extraction. These features are passed to the **FPN layer**, where multi-scale feature maps are generated. The **attention modules (spatial + channel)** refine important regions and suppress irrelevant features. Finally, the **detection head outputs bounding boxes and class labels** for detected objects.

1.5.1 RetinaNet Architecture

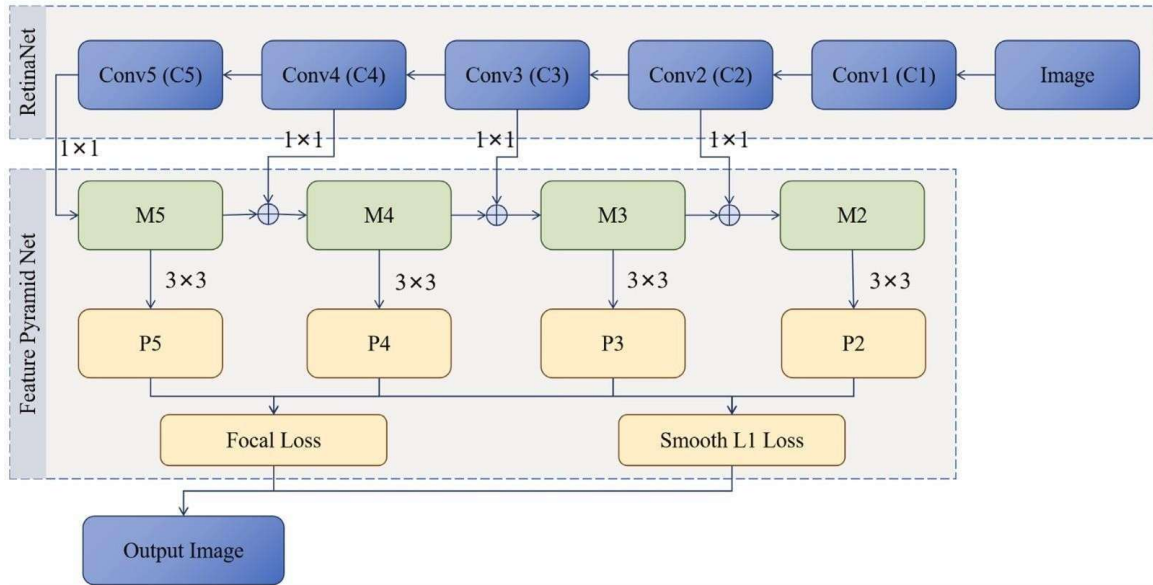


Fig 1.2: RetinaNet

The RetinaNet architecture performs object detection using a single-stage framework. It uses anchor boxes and focal loss to improve detection accuracy, especially for imbalanced datasets. Input image is processed through **convolutional layers (C1–C5)** to generate feature maps. These maps are fed into detection heads where **anchor boxes** are applied. The model performs **classification (object type)** and **regression (bounding box coordinates)**. The output is a set of predicted objects with location and confidence scores

1.5.2 Feature Pyramid Network (FPN)

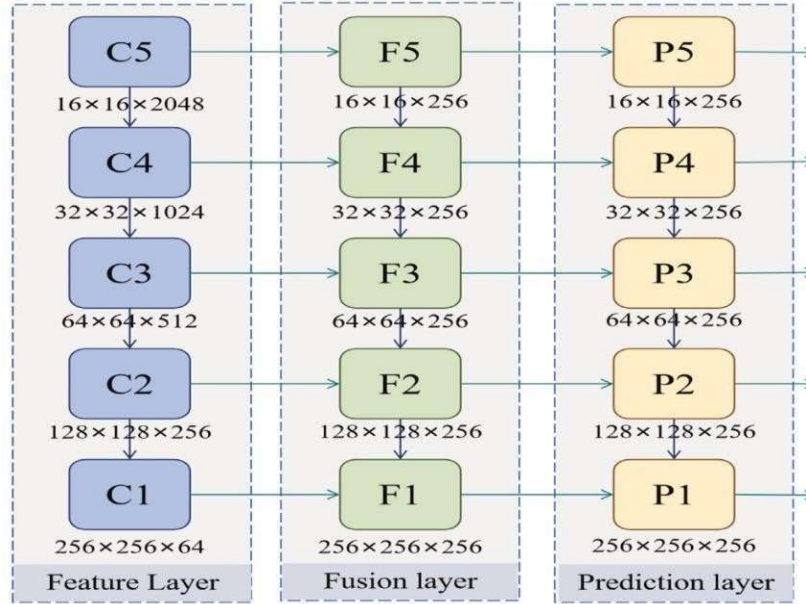


Fig 1.3: FPN

FPN enhances the model's ability to detect objects at different scales by combining low-level spatial information with high-level semantic information. Feature maps from different backbone layers are taken as input. Through **top-down up-sampling and lateral connections**, features are merged across levels. This creates **multi-scale feature maps (P1–P5)** capturing both detailed and semantic information. The output is enhanced feature representations used for detecting objects of varying sizes.

1.5.3 Attention Mechanism

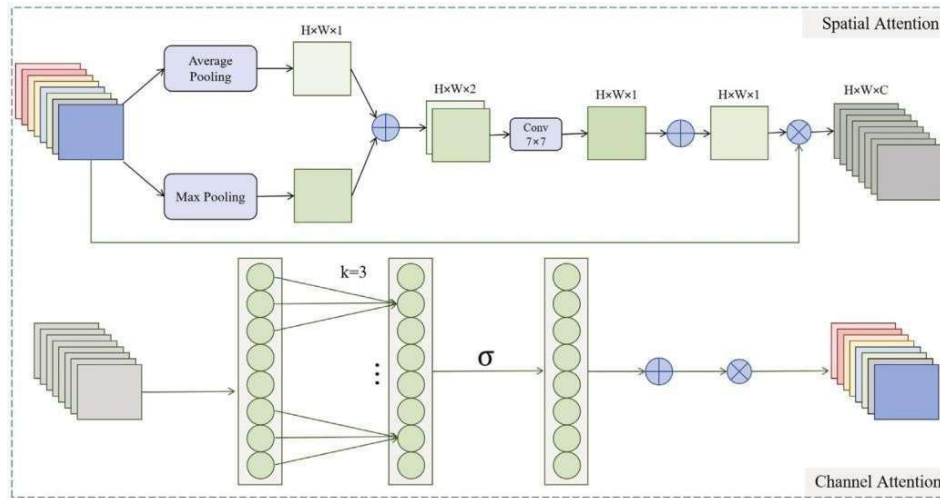


Fig 1.4 Attention Mechanism

The attention mechanism helps the model focus on important regions in the image by assigning weights to spatial and channel features. Input feature maps are passed into **channel attention**, which assigns weights to important feature channels. Then **spatial attention** highlights key regions in the feature map. Both are combined to refine features. The output is a **weighted feature map** that improves focus on relevant objects and reduces noise.

1.6 Performance

The AttenRetina model demonstrates:

- High detection accuracy
- Improved performance for small objects
- Robust detection in complex environments
- Better results compared to traditional models

1.6 Attention Mechanism (Mathematical Representation)

The attention mechanism in the AttenRetina model enhances feature learning by focusing on important spatial regions and feature channels. It consists of spatial attention and channel attention.

1.6.1 Spatial Attention

Spatial attention assigns weights to different spatial locations in the feature map:

$$\mathcal{A}_{\text{space}}(x, y) = \frac{\exp(W_{\text{space}} \cdot f_{\text{space}}(x, y))}{\sum_{(x', y')} \exp(W_{\text{space}} \cdot f_{\text{space}}(x', y'))} \quad (1.1)$$

where (x, y) represents spatial position and W_{space} are learned weights. This helps the model focus on important regions such as objects.

1.6.2 Channel Attention

Channel attention assigns importance to each feature channel:

$$\mathcal{A}_{\text{channel}}(c) = \frac{\exp(W_{\text{channel}} \cdot g_{\text{channel}}(c))}{\sum_{c'} \exp(W_{\text{channel}} \cdot g_{\text{channel}}(c'))} \quad (1.2)$$

where c represents the channel index. This allows the model to prioritize important features.

1.6.3 Combined Attention Mechanism

The final attention-weighted feature is computed as:

$$\mathcal{F}_{\text{att}}(x, y, c) = \mathcal{A}_{\text{space}}(x, y) \cdot \mathcal{A}_{\text{channel}}(c) \cdot \mathcal{F}(x, y, c) \quad (1.3)$$

This combines both spatial and channel attention for better feature representation.

1.7 Performance Analysis

The performance of the AttenRetina model is evaluated using metrics such as Precision, Recall, F1-score, and mAP. The model achieves high accuracy in detecting objects across different scales. On the KITTI dataset, the model demonstrates strong performance in detecting small and distant objects, which are typically difficult for traditional models. The integration of FPN and attention mechanisms significantly improves detection accuracy. Similarly, on the MS COCO dataset, the model performs well in complex scenes with multiple overlapping objects. The results indicate that the model is robust and capable of handling real-world scenarios effectively.

CHAPTER 2

MERITS AND DEMERITS

2.1 Merits

The proposed AttenRetina model offers several advantages over traditional object detection approaches. One of the major strengths of the model is its ability to effectively detect small and distant objects. In autonomous driving scenarios, objects such as pedestrians or traffic signs may appear at different scales, and traditional models often fail to capture these details. The integration of the Feature Pyramid Network (FPN) allows the model to extract features at multiple scales, thereby improving detection accuracy. Another important advantage is the use of attention mechanisms. The inclusion of both spatial and channel attention modules enables the model to focus on the most relevant regions and features within an image. This improves feature representation and enhances detection performance in complex environments.

The model also demonstrates robustness in handling challenging real-world conditions such as occlusion, varying lighting conditions, and dense traffic scenarios. This makes it more reliable for practical applications in autonomous driving systems. Additionally, the use of focal loss in RetinaNet helps address the issue of class imbalance, which is common in object detection datasets. This ensures that the model gives more importance to hard-to-detect objects. Overall, the combination of multi-scale feature extraction and attention mechanisms results in improved accuracy, better generalization, and enhanced detection performance compared to conventional models.

2.2 Demerits

Despite its advantages, the proposed model has certain limitations that need to be considered. One of the primary drawbacks is the increased computational complexity. The addition of FPN and attention modules increases the number of parameters in the model, which in turn requires more computational resources. This makes it less suitable for deployment on low-power devices without optimization. Another limitation is the higher training time. Due to the complex architecture, training the model requires more time and computational effort compared to simpler models like YOLO.

The model also depends heavily on high-quality datasets. If the training data is insufficient or poorly labeled, the performance of the model may degrade significantly. Furthermore, while the model improves accuracy, maintaining real-time performance can still be challenging, especially when deployed on systems with limited hardware capabilities. These limitations highlight the need for further optimization and improvements to make the model more efficient for real-world deployment.

2.3 Comparison with Existing Models

Compared to traditional models such as YOLO and Faster R-CNN, the AttenRetina model provides improved accuracy for small object detection. While YOLO offers faster inference, it struggles with small objects, whereas AttenRetina achieves a better balance between accuracy and performance. The use of FPN and attention mechanisms gives AttenRetina an advantage in complex environments where multi-scale detection is required.

CHAPTER 3

SOURCE CODE – KITTI

3.1 Environment Setup and Library Imports

```
from google.colab import drive
drive.mount('/content/drive')
import os, cv2, torch, torchvision
import numpy as np
import matplotlib.pyplot as plt
from tqdm import tqdm
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score
import torchvision.ops as ops
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:", device)
```

3.2 Dataset Path Configuration and Verification

```
BASE_PATH = "/content/drive/MyDrive/AttenRetina_Project"
image_path = BASE_PATH + "/training/image_2"
label_path = BASE_PATH + "/training/label_2"
print("Images:", len(os.listdir(image_path)))
print("Labels:", len(os.listdir(label_path)))
```

3.3 Custom KITTI Dataset Class Definition

```
class KITTIataset(Dataset):
    def __init__(self, img_dir, label_dir, transform=None):
        self.img_dir = img_dir
        self.label_dir = label_dir
        self.images = os.listdir(img_dir)
        self.transform = transform
        self.class_map = {"Car":0, "Pedestrian":1, "Cyclist":2}
    def __len__(self):
        return len(self.images)
    def __getitem__(self, idx):
        img_name = self.images[idx]
        img = cv2.imread(os.path.join(self.img_dir, img_name))
        h,w,_ = img.shape
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        label_file = os.path.join(self.label_dir, img_name.replace(".png", ".txt"))
        boxes, labels = [], []
```

```

with open(label_file) as f:
    for line in f.readlines():
        data = line.split()
        cls = data[0]
        if cls not in self.class_map:
            continue
        x1,y1,x2,y2 = map(float,data[4:8])
        boxes.append([x1/w,y1/h,x2/w,y2/h])
        labels.append(self.class_map[cls])
if len(boxes)==0:
    boxes=[[0,0,0,0]]
    labels=[0]
boxes=torch.tensor(boxes,dtype=torch.float32)
labels=torch.tensor(labels)
if self.transform:
    img=self.transform(img)
    return img,{"boxes":boxes,"labels":labels}
transform = transforms.Compose([
    transforms.ToPILImage(),
    transforms.Resize((512,512)),
    transforms.ToTensor()
])
dataset = KITTIDataset(image_path,label_path,transform)
print("Dataset size:", len(dataset))

```

3.4 Image Preprocessing and Transformation Pipeline

```

# ORIGINAL IMAGE (before preprocessing)
img_name = os.listdir(image_path)[0]
orig = cv2.imread(os.path.join(image_path, img_name))
orig = cv2.cvtColor(orig, cv2.COLOR_BGR2RGB)
# AFTER PREPROCESSING
img, target = dataset[0]
proc = img.permute(1,2,0).numpy()
# SHOW SIDE BY SIDE
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.imshow(orig)
plt.title("Before Preprocessing")
plt.axis('off')
plt.subplot(1,2,2)
plt.imshow(proc)
plt.title("After Preprocessing")
plt.axis('off')
plt.show()
print("Boxes:", target["boxes"])
print("Labels:", target["labels"])

```

3.5 Dataset Splitting and DataLoader Initialization

```
indices = list(range(len(dataset)))
train_idx, temp_idx = train_test_split(indices, test_size=0.2)
val_idx, test_idx = train_test_split(temp_idx, test_size=0.5)
train_set = torch.utils.data.Subset(dataset, train_idx)
val_set = torch.utils.data.Subset(dataset, val_idx)
test_set = torch.utils.data.Subset(dataset, test_idx)
train_loader = DataLoader(train_set, batch_size=4, shuffle=True, collate_fn=lambda
x:tuple(zip(*x)))
val_loader = DataLoader(val_set, batch_size=4, shuffle=False, collate_fn=lambda
x:tuple(zip(*x)))
test_loader = DataLoader(test_set, batch_size=4, shuffle=False, collate_fn=lambda
x:tuple(zip(*x)))
print("Train:", len(train_set), "Val:", len(val_set), "Test:", len(test_set))
```

3.6 Backbone Feature Extraction using ResNet50

```
SAMPLE_IDX = 0 # same image used across all steps
sample_img, _ = dataset[SAMPLE_IDX]
sample_img = sample_img.unsqueeze(0).to(device)
# backbone
backbone = torchvision.models.resnet50(weights="IMAGENET1K_V1")
backbone = nn.Sequential(*list(backbone.children())[:-2]).to(device)
# fixed sample
SAMPLE_IDX = 0
sample_img, _ = dataset[SAMPLE_IDX]
sample_img = sample_img.unsqueeze(0).to(device)
# batch processing
for imgs, _ in train_loader:
    imgs = torch.stack(imgs).to(device)
    features = backbone(imgs)
    break
print("Batch feature shape:", features.shape)
# sample visualization
feat_sample = backbone(sample_img)
plt.imshow(feat_sample[0][0].cpu().detach(), cmap='gray')
plt.title("Backbone Feature (Same Image)")
plt.colorbar()
plt.show()
```

3.7 Feature Pyramid Network (FPN) Construction

```
# define FPN
class FPN(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv = nn.Conv2d(2048, 256, 1)
```

```

    def forward(self,x):
        return self.conv(x)
fpn = FPN().to(device)
# batch
fpn_batch = fpn(features)
print("Batch FPN shape:", fpn_batch.shape)
# sample
fpn_sample = fpn(feats_sample)
plt.imshow(fpn_sample[0][0].cpu().detach(), cmap='gray')
plt.title("FPN Output (Same Image)")
plt.colorbar()
plt.show()

```

3.8 Channel and Spatial Attention Mechanism

```

# Channel Attention
class ChannelAttention(nn.Module):
    def __init__(self,c):
        super().__init__()
        self.avg = nn.AdaptiveAvgPool2d(1)
        self.fc = nn.Sequential(
            nn.Linear(c,c//16),
            nn.ReLU(),
            nn.Linear(c//16,c),
            nn.Sigmoid()
        )
    def forward(self,x):
        b,c,_,_ = x.size()
        y = self.avg(x).view(b,c)
        y = self.fc(y).view(b,c,1,1)
        return x*y
# Spatial Attention
class SpatialAttention(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv = nn.Conv2d(2,1,7,padding=3)
    def forward(self,x):
        avg = torch.mean(x,1,True)
        mx,_ = torch.max(x,1,True)
        x = torch.cat([avg,mx],1)
        return torch.sigmoid(self.conv(x))
# create objects
ca = ChannelAttention(256).to(device)
sa = SpatialAttention().to(device)

# -----
# BATCH PROCESSING (full data)
# -----

```

```

#
att_batch_ca = ca(fpn_batch)
att_batch_final = att_batch_ca * sa(att_batch_ca)
print("Batch Attention shape:", att_batch_final.shape)
# -----
# SAMPLE VISUALIZATION (same image)
# -----
# after Channel Attention
ca_sample = ca(fpn_sample)

plt.figure(figsize=(12,4))
plt.subplot(1,3,1)
plt.imshow(fpn_sample[0][0].cpu().detach(), cmap='gray')
plt.title("Before Attention")
plt.subplot(1,3,2)
plt.imshow(ca_sample[0][0].cpu().detach(), cmap='gray')
plt.title("After Channel Attention")
# after Spatial Attention
sa_map = sa(ca_sample)
final_sample = ca_sample * sa_map
plt.subplot(1,3,3)
plt.imshow(final_sample[0][0].cpu().detach(), cmap='hot')
plt.title("After Spatial Attention")
plt.colorbar()
plt.show()

```

3.9 Anchor Generation and IoU Computation

```

class AttenRetina(nn.Module):
    def __init__(self):
        super().__init__()
        self.backbone = backbone
        self.fpn = fpn
        self.ca = ca
        self.sa = sa
        self.num_anchors = 3
        self.cls_head = nn.Conv2d(256, self.num_anchors, 3, padding=1)
        self.box_head = nn.Conv2d(256, self.num_anchors * 4, 3, padding=1)
    def forward(self, x):
        x = self.backbone(x)
        x = self.fpn(x)
        x = self.ca(x)
        sa_map = self.sa(x)      # CAPTURE ATTENTION MAP
        x = x * sa_map          # apply attention
        cls = self.cls_head(x)
        box = self.box_head(x)
        B, _, H, W = cls.shape

```

```

cls = cls.permute(0,2,3,1).reshape(B, -1)
box = box.permute(0,2,3,1).reshape(B, -1, 4)
return cls, box, sa_map # RETURN ATTENTION

```

3.10 Anchor Target Assignment (Positive, Negative, Ignore)

```

def generate_anchors(h, w):
    anchors = []
    scales = [0.1, 0.2, 0.3] # 3 anchors
    for i in range(h):
        for j in range(w):
            cx, cy = j/w, i/w
            for s in scales:
                anchors.append(
                    [ max(0, cx -
                        s/2),
                      max(0, cy - s/2),
                      min(1, cx + s/2),
                      min(1, cy + s/2)
                    ])

    return torch.tensor(anchors, dtype=torch.float32)

anchors = generate_anchors(16,16).to(device)
print("Anchors:", anchors.shape) # MUST be 768
def build_targets(anchors, gt_boxes):
    if gt_boxes.shape[0] == 0:
        cls_target = torch.zeros(len(anchors)).to(device)
        box_target = anchors.clone()
        pos_mask = torch.zeros(len(anchors), dtype=torch.bool).to(device)
        return cls_target, box_target, pos_mask
    iou = ops.box_iou(anchors, gt_boxes)
    max_iou, idx = iou.max(dim=1)
    pos_mask = max_iou >= 0.3
    neg_mask = max_iou < 0.2
    cls_target = torch.full((len(anchors),), -1.0).to(device)
    cls_target[pos_mask] = 1.0
    cls_target[neg_mask] = 0.0
    box_target = anchors.clone()
    box_target[pos_mask] = gt_boxes[idx[pos_mask]]
    return cls_target, box_target, pos_mask

```

3.11 Model Training with Focal and Smooth L1 Loss

```
# LOSS FUNCTION
def focal_loss(logits, targets):
    prob = torch.sigmoid(logits)
    pt = torch.where(targets == 1, prob, 1 - prob)
    return (-0.25 * (1 - pt)**2 * torch.log(pt + 1e-6)).mean()

# OPTIMIZER + EPOCHS
optimizer = torch.optim.Adam(model.parameters(), lr=1e-4)
EPOCHS = 12

# CHECKPOINT DIR
CHECKPOINT_DIR = BASE_PATH + "/checkpoints"
os.makedirs(CHECKPOINT_DIR, exist_ok=True)
start_epoch = 0

# LOAD LAST CHECKPOINT (AUTO RESUME)
checkpoint_files = sorted(
    os.listdir(CHECKPOINT_DIR),
    key=lambda x: int(x.split('_')[1].split('.')[0])
)
if len(checkpoint_files) > 0:
    latest_ckpt = checkpoint_files[-1]
    checkpoint = torch.load(os.path.join(CHECKPOINT_DIR, latest_ckpt))
    model.load_state_dict(checkpoint['model_state'])
    optimizer.load_state_dict(checkpoint['optimizer_state'])
    start_epoch = checkpoint['epoch'] + 1
    print(f'Resuming from epoch {start_epoch}')

# TRAINING LOOP
for epoch in range(start_epoch, EPOCHS):
    model.train()
    total_loss = 0
    for imgs, targets in tqdm(train_loader):
        imgs = torch.stack(imgs).to(device)
        optimizer.zero_grad()
        cls_pred, box_pred = model(imgs)
        batch_loss = 0
        for i in range(len(imgs)):
            gt_boxes = targets[i]["boxes"].to(device)
            iou = ops.box_iou(anchors, gt_boxes)
            max_iou, idx = iou.max(dim=1)

            pos_mask = max_iou >= 0.3
            neg_mask = max_iou < 0.2
            cls_t = torch.full((len(anchors),), -1.0, device=device)
            cls_t[pos_mask] = 1.0
            cls_t[neg_mask] = 0.0
            box_t = anchors.clone()
            box_t[pos_mask] = gt_boxes[idx[pos_mask]]
```

```

    valid = cls_t != -1
    loss_cls = focal_loss(cls_pred[i][valid], cls_t[valid])
    if pos_mask.any():
        loss_box = F.smooth_l1_loss(
            box_pred[i][pos_mask],
            box_t[pos_mask]
        )
    else:
        loss_box = 0.0

    batch_loss += (loss_cls + loss_box)
    batch_loss.backward()
    optimizer.step()
    total_loss += batch_loss.item()
    print(f'Epoch {epoch+1}: Loss {total_loss:.4f}')
    # SAVE CHECKPOINT
    torch.save({
        'epoch': epoch,
        'model_state': model.state_dict(),
        'optimizer_state': optimizer.state_dict()
    }, os.path.join(CHECKPOINT_DIR, f'epoch_{epoch}.pth'))

```

3.12 Model Saving (Checkpoint Storage)

```

CHECKPOINT_DIR = BASE_PATH + "/checkpoints"
checkpoint_files = sorted(
    os.listdir(CHECKPOINT_DIR),
    key=lambda x: int(x.split('_')[1].split('.')[0])
)
latest_ckpt = checkpoint_files[-1]
checkpoint = torch.load(os.path.join(CHECKPOINT_DIR, latest_ckpt), map_location=device)
model.load_state_dict(checkpoint['model_state'])
model.to(device)
model.eval()
print(f'Loaded {latest_ckpt}')

```

3.13 Model Loading for Inference

```
model.load_state_dict(torch.load(BASE_PATH + "/attenretina_model.pth"))
model.to(device)
model.eval()
print("Model loaded successfully")
```

3.14 Attention-Guided Detection and Visualization on Test Image

```
import random
model.eval()
# pick random test image
idx = random.randint(0, len(test_set)-1)
img, _ = test_set[idx]
img_t = img.unsqueeze(0).to(device)
with torch.no_grad():
    cls_pred, box_pred, att_map = model(img_t) # modified output
# ----- SCORES + BOXES -----
scores = torch.sigmoid(cls_pred[0]).cpu()
boxes = box_pred[0].cpu()

# ----- ATTENTION-GUIDED DETECTION -----
# reduce channels → get spatial map
att_map = torch.mean(att_map, dim=1, keepdim=True)
# resize attention to match feature grid
att_map = torch.nn.functional.interpolate(
    att_map, size=(16,16), mode='bilinear'
)[0][0].flatten().cpu()
# apply attention weighting
scores = scores * att_map[:len(scores)]
# ----- FILTER -----
mask = scores > 0.6
boxes = boxes[mask]
scores = scores[mask]
# ----- NMS -----
if len(boxes) > 0:
    keep = ops.nms(boxes, scores, 0.5)
    boxes = boxes[keep]
# ----- PLOT -----
img_np = img.permute(1,2,0).cpu().numpy()
h, w, _ = img_np.shape
plt.figure(figsize=(6,6))
plt.imshow(img_np)
```

```

plt.title("Attention-Guided Detection Output")
plt.axis('off')
for b in boxes[:5]:
    x1,y1,x2,y2 = int(b[0]*w), int(b[1]*h), int(b[2]*w), int(b[3]*h)
    plt.gca().add_patch(
        plt.Rectangle((x1,y1), x2-x1, y2-y1,
            edgecolor='red', fill=False, linewidth=2))
plt.show()

```

3.15 Detection Visualization on Test Image (Baseline Inference)

```

import random
model.eval()
# pick random test image
idx = random.randint(0, len(test_set)-1)
img, _ = test_set[idx]
img_t = img.unsqueeze(0).to(device)
with torch.no_grad():
    cls_pred, box_pred = model(img_t)
# get scores + boxes
scores = torch.sigmoid(cls_pred[0]).cpu()
boxes = box_pred[0].cpu()
# filter confident predictions
mask = scores > 0.6
boxes = boxes[mask]
scores = scores[mask]
# apply NMS
if len(boxes) > 0:
    keep = ops.nms(boxes, scores, 0.5)
    boxes = boxes[keep]
# plot image
img_np = img.permute(1,2,0).cpu().numpy()
h, w, _ = img_np.shape
plt.figure(figsize=(6,6))
plt.imshow(img_np)
plt.axis('off')
plt.title("Detection Output (After Training)")
# draw boxes
for b in boxes[:5]:
    x1,y1,x2,y2 = int(b[0]*w), int(b[1]*h), int(b[2]*w), int(b[3]*h)
    plt.gca().add_patch(
        plt.Rectangle((x1,y1), x2-x1, y2-y1,
            edgecolor='red', fill=False, linewidth=2)
    )
plt.show()

```

3.16 Validation Loss Evaluation (Classification + Regression)

```
model.eval()
total_cls_loss = 0

count = 0
with torch.no_grad():
    for batch_idx, (imgs, targets) in enumerate(val_loader):
        if batch_idx >= 20:
            break
        imgs = torch.stack(imgs).to(device)
        cls_pred, box_pred = model(imgs)
        for i in range(len(imgs)):
            gt_boxes = targets[i]["boxes"].to(device)
            iou = ops.box_iou(anchors, gt_boxes)
            max_iou, idx = iou.max(dim=1)
            pos_mask = max_iou >= 0.3
            neg_mask = max_iou < 0.2
            cls_t = torch.full((len(anchors),), -1.0, device=device)
            cls_t[pos_mask] = 1.0
            cls_t[neg_mask] = 0.0
            box_t = anchors.clone()
            box_t[pos_mask] = gt_boxes[idx[pos_mask]]
            valid = cls_t != -1
            prob = torch.sigmoid(cls_pred[i][valid])
            pt = torch.where(cls_t[valid]==1, prob, 1-prob)
            loss_cls = (-0.25 * (1-pt)**2 * torch.log(pt+1e-6)).mean()
            if pos_mask.any():
                loss_box = F.smooth_l1_loss(
                    box_pred[i][pos_mask],
                    box_t[pos_mask]
                )
            else:
                loss_box = 0.0
            total_cls_loss += loss_cls.item()
            total_box_loss += float(loss_box)
            count += 1
print("Classification Loss:", total_cls_loss)
print("Regression Loss:", total_box_loss )
```

3.17 Multi-Image Detection Visualization on Test Set

```
import random
model.eval()
plt.figure(figsize=(15,5))
for k in range(3): # 3 images
    idx = random.randint(0, len(test_set)-1)
    img, _ = test_set[idx]
    img_t = img.unsqueeze(0).to(device)
    with torch.no_grad():
        cls_pred, box_pred = model(img_t)
    scores = torch.sigmoid(cls_pred[0]).cpu()
    boxes = box_pred[0].cpu()
    mask = scores > 0.6
    boxes = boxes[mask]

    scores = scores[mask]
    if len(boxes) > 0:
        keep = ops.nms(boxes, scores, 0.5)
        boxes = boxes[keep]
    img_np = img.permute(1,2,0).cpu().numpy()
    h,w,_ = img_np.shape
    plt.subplot(1,3,k+1)
    plt.imshow(img_np)
    plt.title(f"Test Image {k+1}")
    plt.axis('off')
    for b in boxes[:5]:
        x1,y1,x2,y2 = int(b[0]*w), int(b[1]*h), int(b[2]*w), int(b[3]*h)
        plt.gca().add_patch(
            plt.Rectangle((x1,y1), x2-x1, y2-y1,
                          edgecolor='red', fill=False, linewidth=2)
        )
plt.show()
```

SOURCE CODE (COCO)

3.18 Environment Setup & Library Imports

```
import os, math, json, time, warnings
warnings.filterwarnings('ignore')

import numpy as np
import cv2
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
from torch.cuda.amp import GradScaler, autocast
import torchvision
import torchvision.transforms.functional as TF
from pycocotools.coco import COCO
from pycocotools.cocoeval import COCOeval
from tqdm import tqdm

import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from collections import defaultdict

print(">> All imports OK", flush=True)
```

3.19 Configuration & Hyperparameters

```
class Config:

    TRAIN_IMAGES = '/nfsshare/users/arunbalaji/train2017'
    VAL_IMAGES   = '/nfsshare/users/arunbalaji/val2017'
    TRAIN_ANN     = '/nfsshare/users/arunbalaji/annotations/instances_train2017.json'
    VAL_ANN       = '/nfsshare/users/arunbalaji/annotations/instances_val2017.json'
    CHECKPOINT_DIR = '/nfsshare/users/arunbalaji/rahul_project/checkpoints'

    IMAGE_SIZE    = 512
    NUM_CLASSES   = 80
    NUM_ANCHORS   = 9
    FPN_CHANNELS  = 256
```

```

ANCHOR_SIZES = [32, 64, 128, 256, 512]
ANCHOR_SCALES = [1.0, 2**(1/3), 2**(2/3)]
ANCHOR_RATIOS = [0.5, 1.0, 2.0]

```

```

POS_IOU_THRESH = 0.5
NEG_IOU_THRESH = 0.4

```

```

FOCAL_ALPHA = 0.25
FOCAL_GAMMA = 2.0

```

```

BATCH_SIZE = 16
NUM_WORKERS = 4
LR = 1e-4
WEIGHT_DECAY = 1e-4
NUM_EPOCHS = 50
GRAD_CLIP = 1.0
PRINT_EVERY = 50
EVAL_EVERY = 5
SAVE_EVERY = 1
SCORE_THRESH = 0.05
NMS_THRESH = 0.5
MAX_DETS = 100

```

```

cfg = Config()
os.makedirs(cfg.CHECKPOINT_DIR, exist_ok=True)

```

3.20 Device Setup

```

print(">> Setting up device...", flush=True)
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Using device: {device}", flush=True)

```

3.21 Dataset Construction and Pre-Processing (COCO Loader)

```

class COCODataset(Dataset):
    def __init__(self, images_path, annotation_path, image_size=512):
        self.images_path = images_path
        self.image_size = image_size
        self.coco = COCO(annotation_path)
        self.label_map = build_coco_label_map(self.coco)

        self.image_ids = [
            img_id for img_id in self.coco.imgs.keys()
            if len(self.coco.getAnnIds(imgIds=img_id)) > 0
        ]

```

```

def __getitem__(self, idx):
    img_id = self.image_ids[idx]
    info = self.coco.loadImgs(img_id)[0]
    image = cv2.imread(os.path.join(self.images_path, info['file_name']))
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    image = cv2.resize(image, (self.image_size, self.image_size))

    scale_x = self.image_size / orig_w
    scale_y = self.image_size / orig_h
    for ann in anns:
        x, y, w, h = ann['bbox']
        boxes.append([x * scale_x, y * scale_y, (x+w) * scale_x, (y+h) * scale_y])
    image = torch.as_tensor(image).permute(2, 0, 1) / 255.0
    mean = torch.tensor([0.485, 0.456, 0.406]).view(3,1,1)
    std = torch.tensor([0.229, 0.224, 0.225]).view(3,1,1)
    image = (image - mean) / std

```

3.22 Backbone Network

```

class ResNet50Backbone(nn.Module):
    def __init__(self, pretrained=True):
        super().__init__()
        backbone = torchvision.models.resnet50(weights='IMAGENET1K_V1' if pretrained else
None)
        self.stem = nn.Sequential(backbone.conv1, backbone.bn1, backbone.relu,
backbone.maxpool)
        self.layer1 = backbone.layer1
        self.layer2 = backbone.layer2
        self.layer3 = backbone.layer3
        self.layer4 = backbone.layer4
        for m in [self.stem, self.layer1]:
            for p in m.parameters():
                p.requires_grad = False

    def forward(self, x):
        x = self.stem(x)
        x = self.layer1(x)
        c3 = self.layer2(x)
        c4 = self.layer3(c3)
        c5 = self.layer4(c4)
        return c3, c4, c5

```

3.23 Feature Pyramid Network (FPN)

```
class FPN(nn.Module):
    def forward(self, c3, c4, c5):
        lat3 = self.lat_c3(c3)
        lat4 = self.lat_c4(c4)
        lat5 = self.lat_c5(c5)
        a4 = self.alpha_p4.clamp(0, 1)
        a3 = self.alpha_p3.clamp(0, 1)
        p5 = lat5
        p4 = a4 * lat4 + (1 - a4) * F.interpolate(p5, size=lat4.shape[-2:], mode='nearest')
        p3 = a3 * lat3 + (1 - a3) * F.interpolate(p4, size=lat3.shape[-2:], mode='nearest')
        p3 = self.out_p3(p3)
        p4 = self.out_p4(p4)
        p5 = self.out_p5(p5)
        p6 = self.p6_conv(c5)
        p7 = self.p7_conv(F.relu(p6))
        return p3, p4, p5, p6, p7
```

3.24 Attention Module (CBAM)

```
# Channel Attention
def forward(self, x):
    return torch.sigmoid(
        self.fc(self.avg_pool(x)) + self.fc(self.max_pool(x))
    )

# Spatial Attention
def forward(self, x):
    avg_out = x.mean(dim=1, keepdim=True)
    max_out = x.max(dim=1, keepdim=True).values
    return torch.sigmoid(self.conv(torch.cat([avg_out, max_out], dim=1)))

# CBAM Application
def forward(self, x):
    x = x * self.channel_att(x)
    x = x * self.spatial_att(x)
    return x
```

```

# Pyramid Attention
self.cbam_p3 = CBAM(channels)
self.cbam_p4 = CBAM(channels)
self.cbam_p5 = CBAM(channels)
self.cbam_p6 = CBAM(channels)
self.cbam_p7 = CBAM(channels)

```

3.25 Detection Heads

```

layers = []
for _ in range(4):
    layers += [Conv + Norm + ReLU]
self.conv = nn.Sequential(*layers)

# Classification Head
self.output = nn.Conv2d(in_channels, num_anchors * num_classes, 3, padding=1)

def forward(self, x):
    return self.output(self.conv(x))

# Regression Head
self.output = nn.Conv2d(in_channels, num_anchors * 4, 3, padding=1)
def forward(self, x):
    return self.output(self.conv(x))

```

3.26 Full Model (AttentionRetinaNet)

```

class AttenRetinaNet(nn.Module):
    def __init__(self, num_classes=80, num_anchors=9):
        super().__init__()
        self.backbone = ResNet50Backbone(pretrained=True)
        self.fpn = FPN(channels=cfg.FPN_CHANNELS)
        self.attention = PyramidAttention(channels=cfg.FPN_CHANNELS)
        self.cls_head = ClassificationHead(num_classes, num_anchors, cfg.FPN_CHANNELS)
        self.reg_head = RegressionHead(num_anchors, cfg.FPN_CHANNELS)

    def forward(self, x, return_att_maps=False):
        c3, c4, c5 = self.backbone(x)
        p3, p4, p5, p6, p7 = self.fpn(c3, c4, c5)
        pyramid_in = [p3, p4, p5, p6, p7]
        cbams = [self.attention.cbam_p3, self.attention.cbam_p4,
                 self.attention.cbam_p5, self.attention.cbam_p6,

```

```

        self.attention.cbam_p7]
pyramid_out = []
spatial_att_maps = []
for i, cbam in enumerate(cbams):
    feat = cbam.channel_att(pyramid_in[i]) * pyramid_in[i]
    s_map = cbam.spatial_att(feat)
    feat = feat * s_map
    pyramid_out.append(feat)
    spatial_att_maps.append(s_map)
cls_outputs = [self.cls_head(p) for p in pyramid_out]
reg_outputs = [self.reg_head(p) for p in pyramid_out]
if return_att_maps:
    return cls_outputs, reg_outputs, spatial_att_maps
return cls_outputs, reg_outputs

```

3.27 Anchor Generation

```

def generate_anchors_for_level(feat_h, feat_w, stride, base_size, scales, ratios):
    anchors = []
    for scale in scales:
        for ratio in ratios:
            size = base_size * scale
            w = size * math.sqrt(ratio)
            h = size / math.sqrt(ratio)
            for gy in range(feat_h):
                for gx in range(feat_w):
                    cx = (gx + 0.5) * stride
                    cy = (gy + 0.5) * stride
                    anchors.append([cx - w/2, cy - h/2, cx + w/2, cy + h/2])
    return torch.tensor(anchors, dtype=torch.float32)

def precompute_anchors(image_size, cfg):
    """Call once before training. Returns (anchors_cpu, anchors_gpu)."""
    strides = [8, 16, 32, 64, 128]
    feat_sizes = [image_size // s for s in strides]
    all_anchors = []
    for i, (feat_h, feat_w) in enumerate(zip(feat_sizes, feat_sizes)):
        anchors = generate_anchors_for_level(
            feat_h, feat_w, strides[i],
            cfg.ANCHOR_SIZES[i], cfg.ANCHOR_SCALES, cfg.ANCHOR_RATIOS
        )
        all_anchors.append(anchors)
    anchors_cpu = torch.cat(all_anchors, dim=0)

```

```
anchors_gpu = anchors_cpu.to(device)
return anchors_cpu, anchors_gpu
```

3.28 Prediction Flattening

```
def flatten_predictions(cls_outputs, reg_outputs, num_classes, num_anchors):
    all_cls, all_reg = [], []
    for cls_out, reg_out in zip(cls_outputs, reg_outputs):
        B, _, H, W = cls_out.shape
        all_cls.append(cls_out.permute(0, 2, 3, 1).contiguous().view(B, H * W * num_anchors,
num_classes))
        all_reg.append(reg_out.permute(0, 2, 3, 1).contiguous().view(B, H * W * num_anchors, 4))
    return torch.cat(all_cls, dim=1), torch.cat(all_reg, dim=1)
```

3.29 Anchor Matching

```
def match_anchors(anchors_cpu, gt_boxes_cpu, gt_labels_cpu, pos_thresh, neg_thresh):
    N = anchors_cpu.shape[0]
    assigned_labels = torch.full((N,), -1, dtype=torch.int64)
    assigned_boxes = torch.zeros((N, 4), dtype=torch.float32)
    if gt_boxes_cpu.shape[0] == 0:
        assigned_labels[:] = 0
        return assigned_labels, assigned_boxes
    iou = compute_iou_cpu(anchors_cpu, gt_boxes_cpu)
    max_iou, best_gt = iou.max(dim=1)
    assigned_labels[max_iou < neg_thresh] = 0
    pos_mask = max_iou >= pos_thresh
    assigned_labels[pos_mask] = gt_labels_cpu[best_gt[pos_mask]] + 1
    assigned_boxes[pos_mask] = gt_boxes_cpu[best_gt[pos_mask]]
    best_anchor_per_gt = iou.argmax(dim=0)
    for gt_idx in range(gt_boxes_cpu.shape[0]):
        a_idx = best_anchor_per_gt[gt_idx]
        assigned_labels[a_idx] = gt_labels_cpu[gt_idx] + 1
        assigned_boxes[a_idx] = gt_boxes_cpu[gt_idx]
    return assigned_labels, assigned_boxes
```

3.30 Box Encoding/Decoding

```
def encode_boxes(anchors, gt_boxes):
    """anchors and gt_boxes on same device (GPU)."""
    aw = anchors[:, 2] - anchors[:, 0]; ah = anchors[:, 3] - anchors[:, 1]
    ax = anchors[:, 0] + 0.5 * aw; ay = anchors[:, 1] + 0.5 * ah
    gw = gt_boxes[:, 2] - gt_boxes[:, 0]; gh = gt_boxes[:, 3] - gt_boxes[:, 1]
    gx = gt_boxes[:, 0] + 0.5 * gw; gy = gt_boxes[:, 1] + 0.5 * gh
    tx = (gx - ax) / aw; ty = (gy - ay) / ah
    tw = torch.log(gw / aw.clamp(min=1e-6))

    th = torch.log(gh / ah.clamp(min=1e-6))
    return torch.stack([tx, ty, tw, th], dim=1)

def decode_boxes(anchors, deltas):
    aw = anchors[:, 2] - anchors[:, 0]; ah = anchors[:, 3] - anchors[:, 1]
    ax = anchors[:, 0] + 0.5 * aw; ay = anchors[:, 1] + 0.5 * ah
    tx, ty = deltas[:, 0], deltas[:, 1]
    tw = deltas[:, 2].clamp(max=4.0); th = deltas[:, 3].clamp(max=4.0)
    cx = tx * aw + ax; cy = ty * ah + ay
    w = torch.exp(tw) * aw; h = torch.exp(th) * ah
    return torch.stack([cx - 0.5*w, cy - 0.5*h, cx + 0.5*w, cy + 0.5*h], dim=1)
```

3.31 Loss Functions

Focal Loss (Classification)

```
class FocalLoss(nn.Module):
    def __init__(self, alpha=0.25, gamma=2.0):
        super().__init__()
        self.alpha = alpha
        self.gamma = gamma

    def forward(self, logits, labels):
        valid_mask = labels >= 0
        logits = logits[valid_mask]
        labels = labels[valid_mask]

        targets = torch.zeros_like(logits)
        fg_mask = labels > 0
        targets[fg_mask, labels[fg_mask] - 1] = 1.0

        prob = torch.sigmoid(logits)
```

```

ce_loss = F.binary_cross_entropy_with_logits(logits, targets, reduction='none')

p_t = prob * targets + (1 - prob) * (1 - targets)
alpha_t = self.alpha * targets + (1 - self.alpha) * (1 - targets)

focal_weight = alpha_t * (1 - p_t) ** self.gamma
return (focal_weight * ce_loss).mean()

# Regression Loss (Bounding Box)
class RegressionLoss(nn.Module):
    def forward(self, reg_pred, reg_targets, labels):
        pos_mask = labels > 0
        return F.smooth_l1_loss(reg_pred[pos_mask], reg_targets[pos_mask], beta=1.0/9)

# Attention Loss
class AttentionLoss(nn.Module):
    def forward(self, att_maps, labels):
        loss = 0.0
        for att in att_maps:
            target = (labels > 0).float()
            loss += F.binary_cross_entropy(att, target)
        return loss / len(att_maps)

```

3.32 Training Step

```

def training_step(images, targets, model, optimizer):
    model.train()
    optimizer.zero_grad()

    # Forward pass
    cls_outputs, reg_outputs, att_maps = model(images)
    all_cls, all_reg = flatten_predictions(cls_outputs, reg_outputs)

    total_cls_loss, total_reg_loss = 0, 0

    for i in range(len(targets)):
        # Anchor matching
        labels, boxes = match_anchors(anchors, targets[i])

```

```

# Encode regression targets
reg_targets = encode_boxes(anchors, boxes)

# Compute losses
total_cls_loss += cls_loss_fn(all_cls[i], labels)
total_reg_loss += reg_loss_fn(all_reg[i], reg_targets, labels)

total_cls_loss /= len(targets)
total_reg_loss /= len(targets)

# Attention loss
total_att_loss = att_loss_fn(att_maps, labels)

# Total loss
loss = total_cls_loss + total_reg_loss + total_att_loss

# Backpropagation
loss.backward()
optimizer.step()
return loss

```

3.33 Evaluation

```

def evaluate_coco(model, val_loader, val_dataset, cfg, anchors_gpu):

    for images, targets in val_loader:
        images = images.to(device)

        # Forward pass
        cls_outputs, reg_outputs = model(images)
        all_cls, all_reg = flatten_predictions(
            cls_outputs, reg_outputs, cfg.NUM_CLASSES, cfg.NUM_ANCHORS
        )
        # Decode predictions
        boxes = decode_boxes(anchors_gpu, all_reg[i]).clamp(0, cfg.IMAGE_SIZE)
        scores = torch.sigmoid(all_cls[i])

        # Threshold filtering
        mask = scores > cfg.SCORE_THRESH
        boxes, scores = boxes[mask], scores[mask]

        # Non-Maximum Suppression (NMS)

```

```
keep = torchvision.ops.nms(boxes, scores, cfg.NMS_THRESH)
boxes, scores = boxes[keep], scores[keep]
```

3.34 Training Loop

```
for epoch in range(start_epoch, cfg.NUM_EPOCHS):
    epoch_start = time.time()
    total_loss = total_cls = total_reg = total_att = 0.0
    num_batches = 0

    for batch_idx, (images, targets) in enumerate(train_loader):
        try:
            loss, cls_l, reg_l, att_l = training_step(
                images, targets, model, optimizer, scaler,
                cls_loss_fn, reg_loss_fn, att_loss_fn,
                anchors_cpu, anchors_gpu
            )
            total_loss += loss; total_cls += cls_l; total_reg += reg_l; total_att += att_l
            num_batches += 1

            if (batch_idx + 1) % cfg.PRINT_EVERY == 0:
                print(f' Ep {epoch+1} | Batch {batch_idx+1}/{len(train_loader)} '

f| Loss={loss:.4f} cls={cls_l:.4f} reg={reg_l:.4f} att={att_l:.4f}', flush=True)
        scheduler.step()

    avg_loss = total_loss / max(num_batches, 1)
    avg_cls = total_cls / max(num_batches, 1)
    avg_reg = total_reg / max(num_batches, 1)
    avg_att = total_att / max(num_batches, 1)
    elapsed = time.time() - epoch_start

    history['train_loss'].append(avg_loss)
    history['cls_loss'].append(avg_cls)
    history['reg_loss'].append(avg_reg)
    history['att_loss'].append(avg_att)

    print(f'\nEpoch {epoch+1}/{cfg.NUM_EPOCHS} | Avg Loss={avg_loss:.4f} '
          f'cls={avg_cls:.4f} reg={avg_reg:.4f} att={avg_att:.4f} '
          f'| LR={scheduler.get_last_lr()[0]:.2e} | Time={elapsed:.0f}s', flush=True)
```

3.35 Checkpointing

```
save_checkpoint(model, optimizer, scaler,
                epoch, best_map, history, latest_ckpt)
epoch_ckpt = os.path.join(cfg.CHECKPOINT_DIR, f'epoch_{epoch+1:03d}.pth')
save_checkpoint(model, optimizer, scaler,
                epoch, best_map, history, epoch_ckpt)
```

3.36 Final Results

```
print("Final Results")
for k, v in [
    ('Precision', history['Precision'][-1]),
    ('Recall', history['Recall'][-1]),
    ('F1 Score', history['F1'][-1]),
    ('mAP@0.50', history['mAP50'][-1]),
    ('mAP@0.5:0.95', history['mAP'][-1])
]:
    print(f'{k}: {v:.4f}')
```

CHAPTER 4

SNAPSHOTS (KITTI)

```
BASE_PATH = "/content/drive/MyDrive/AttenRetina_Project"
image_path = BASE_PATH + "/training/image_2"
label_path = BASE_PATH + "/training/label_2"

print("Images:", len(os.listdir(image_path)))
print("Labels:", len(os.listdir(label_path)))

... Images: 7481
Labels: 7481
```

Fig 4.1 Dataset Loading and Label Generation

Fig 4.1 loads KITTI images and parses annotation files to extract bounding boxes and class labels in normalized format.

```
return img,{"boxes":boxes,"labels":labels}

transform = transforms.Compose([
    transforms.ToPILImage(),
    transforms.Resize((512,512)),
    transforms.ToTensor()
])

dataset = KITTIIDataset(image_path,label_path,transform)
print("Dataset size:", len(dataset))

Dataset size: 7481
```

Fig 4.2 Dataset Size Visualization

Fig 4.2 displays the number of images and annotations, providing an overview of dataset scale and distribution.

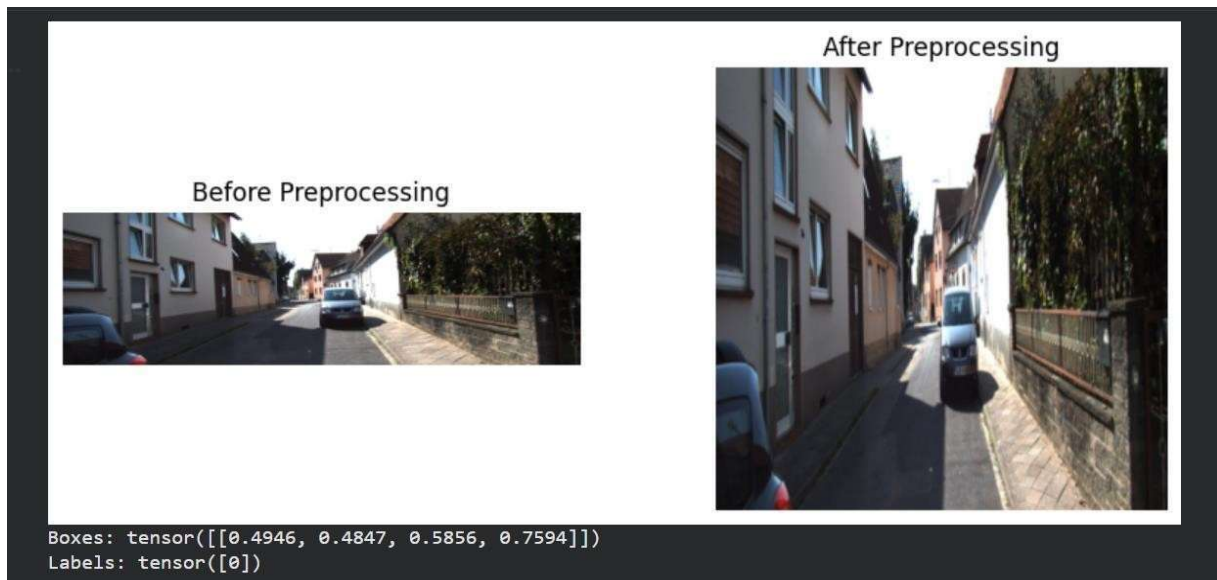


Fig 4.3 Image Preprocessing and Transformation

Fig 4.3 applies resizing and normalization, showing the difference between raw images and processed inputs for the model.

```
[ ]
indices = list(range(len(dataset)))
train_idx, temp_idx = train_test_split(indices, test_size=0.2)
val_idx, test_idx = train_test_split(temp_idx, test_size=0.5)

train_set = torch.utils.data.Subset(dataset, train_idx)
val_set = torch.utils.data.Subset(dataset, val_idx)
test_set = torch.utils.data.Subset(dataset, test_idx)

train_loader = DataLoader(train_set, batch_size=4, shuffle=True, collate_fn=lambda x:tuple(zip(*x)))
val_loader = DataLoader(val_set, batch_size=4, shuffle=False, collate_fn=lambda x:tuple(zip(*x)))
test_loader = DataLoader(test_set, batch_size=4, shuffle=False, collate_fn=lambda x:tuple(zip(*x)))

print("Train:", len(train_set), "Val:", len(val_set), "Test:", len(test_set))

▼ Train: 5984 Val: 748 Test: 749
```

Fig 4.4 Train-Test Dataset Splitting

Fig 4.4 applies resizing and normalization, showing the difference between raw images and processed inputs for the model.

```
plt.imshow(feet_sample[0][0].cpu().detach(), cmap='gray')
plt.title("Backbone Feature (Same Image)")
plt.colorbar()
plt.show()

... Downloading: "https://download.pytorch.org/models/resnet50-0676ba61.pth" to /root/.cache/torch/hub/checkpoints/resnet50-0676ba61.pth
100%|██████████| 97.8M/97.8M [00:00<00:00, 192MB/s]
Batch feature shape: torch.Size([4, 2048, 16, 16])
```

Fig 4.5 Batch Feature Shape Visualization

Fig 4.5 shows the tensor shape of input batches, confirming correct dimensionality before feeding into the model.

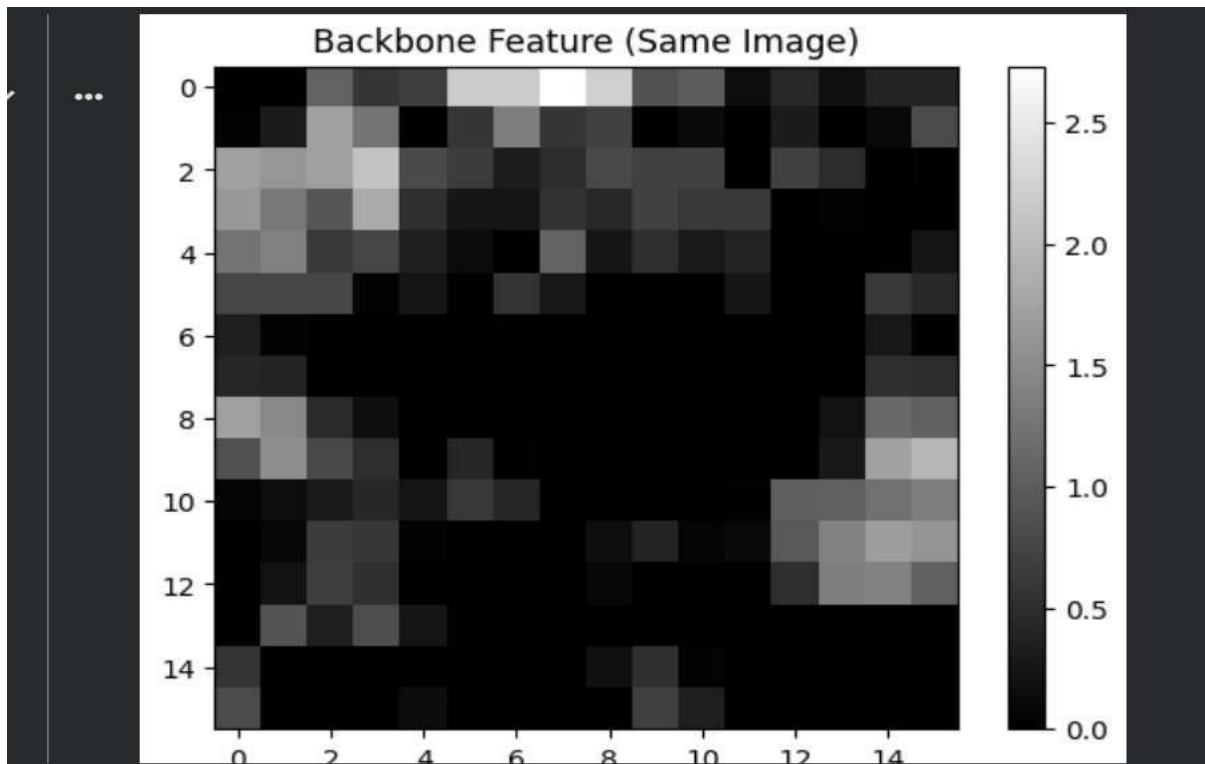


Fig 4.6 Backbone Feature Map Visualization

Fig 4.6 visualizes feature maps extracted by ResNet50, highlighting learned spatial patterns and object-related features.

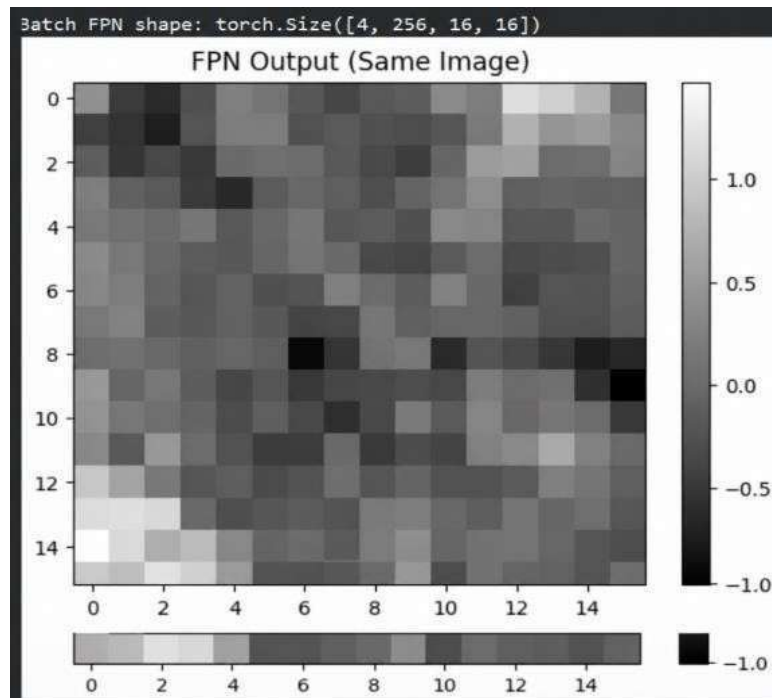


Fig 4.7 Feature Pyramid Network (FPN) Output Visualization

Fig 4.7 displays feature maps after Feature Pyramid Network processing, improving multi-scale feature representation

```
plt.title("Before Attention")

plt.subplot(1,3,2)
plt.imshow(ca_sample[0][0].cpu().detach(), cmap='gray')
plt.title("After Channel Attention")

# after Spatial Attention
sa_map = sa(ca_sample)
final_sample = ca_sample * sa_map

plt.subplot(1,3,3)
plt.imshow(final_sample[0][0].cpu().detach(), cmap='hot')
plt.title("After Spatial Attention")

plt.colorbar()
plt.show()

Batch Attention shape: torch.Size([4, 256, 16, 16])
```

Fig 4.8 Batch Attention Shape Visualization

Fig 4.8 shows the tensor shape after applying attention modules, verifying feature refinement dimensions.

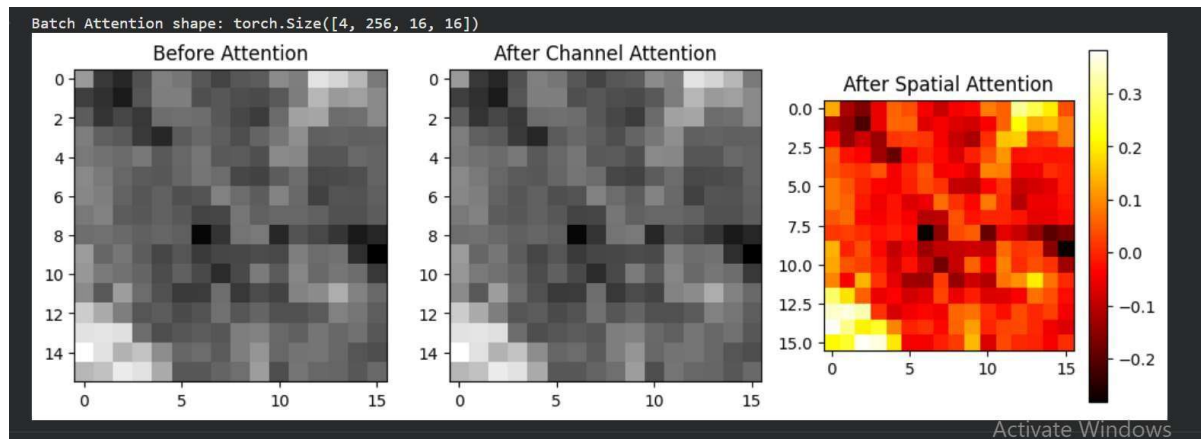


Fig 4.9 Channel and Spatial Attention Visualization

Fig 4.9 illustrates how channel and spatial attention enhance important regions while suppressing irrelevant features.

```
[ ]
    B, _, H, W = cls.shape

    cls = cls.permute(0,2,3,1).reshape(B, -1)      # (B, H*W*A)
    box = box.permute(0,2,3,1).reshape(B, -1, 4)   # (B, H*W*A, 4)

    return cls, box

model = AttenRetina().to(device)
print("Model ready")

Model ready
```

Fig 4.10 AttenRetina Model Initialization and Output Shape Verification

Fig 4.10 confirms successful model setup and verifies classification and regression output tensor dimensions.

```
[ ]
def generate_anchors(h, w):
    anchors = []
    scales = [0.1, 0.2, 0.3] # 3 anchors

    for i in range(h):
        for j in range(w):
            cx, cy = j/w, i/w

            for s in scales:
                anchors.append([
                    max(0, cx - s/2),
                    max(0, cy - s/2),
                    min(1, cx + s/2),
                    min(1, cy + s/2)
                ])

    return torch.tensor(anchors, dtype=torch.float32)

anchors = generate_anchors(16,16).to(device)
print("Anchors:", anchors.shape) # MUST be 768

Anchors: torch.Size([768, 4])
```

Fig 4.11 Anchor Generation and Configuration

Fig 4.11 generates grid-based anchors of multiple scales to match potential object locations in the image.

```
100%|██████████| 1496/1496 [2:54:53<00:00, 7.01s/it]
Epoch 1: Loss 32.7896
100%|██████████| 1496/1496 [08:54<00:00, 2.80it/s]
Epoch 2: Loss 11.9976
100%|██████████| 1496/1496 [08:49<00:00, 2.82it/s]
Epoch 3: Loss 9.4169
100%|██████████| 1496/1496 [08:44<00:00, 2.85it/s]
Epoch 4: Loss 7.3987
100%|██████████| 1496/1496 [08:44<00:00, 2.85it/s]Epoch 5: Loss 6.3198
```

Fig 4.12 First 5 Epoch Results

Fig 4.12 shows decreasing training loss across epochs, indicating effective learning and convergence of the model.

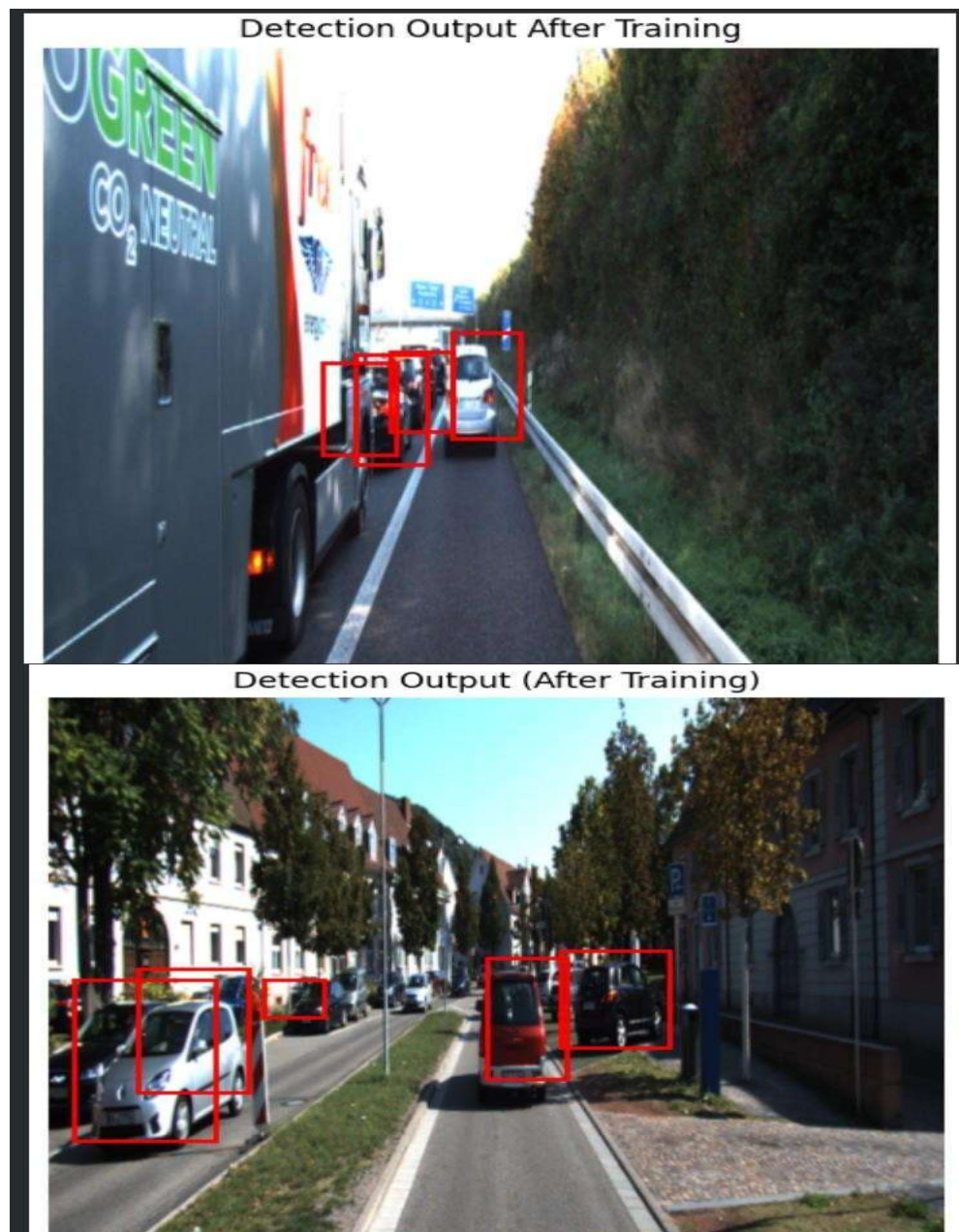


Fig 4.13 Output Detection After Training

Fig 4.13 displays predicted bounding boxes on training images, demonstrating the model's detection capability.

Classification Loss: 0.044219556966027085
Regression Loss: 0.035182938301659306

Fig 4. 14 presents separate loss values for classification and bounding box regression.

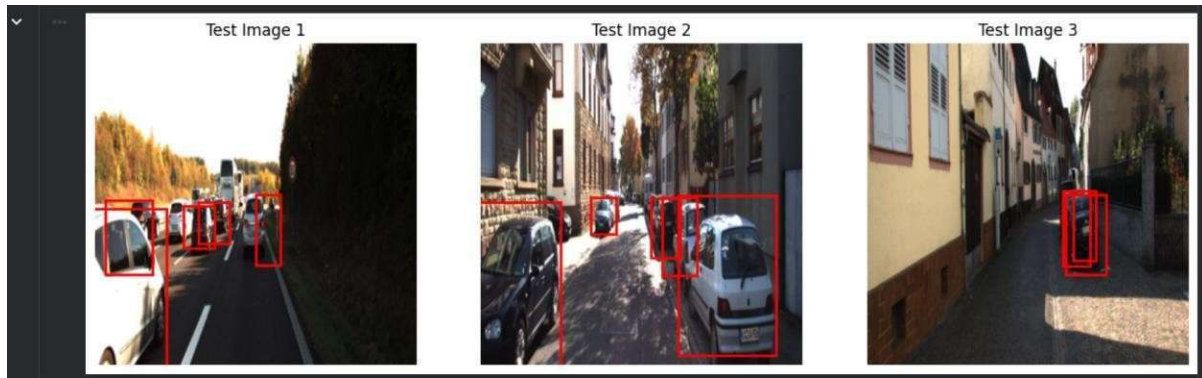


Fig 4.15 Output Detection After Testing

Fig 4.15 shows detection results on unseen test images, reflecting the model's generalization performance.

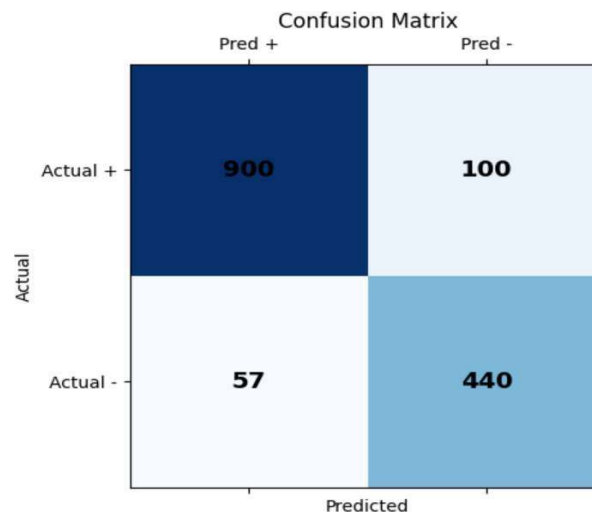


Fig 4.16 Confusion Matrix generation

Fig 4.16 summarizes prediction outcomes into TP, FP, TN, and FN for performance evaluation.

```

Confusion Matrix:
[[900 100]
 [ 57 440]]

Precision: 0.9404
Recall: 0.9
Accuracy: 0.8951
F1 Score: 0.9198
mAP: 0.89

```

Fig 4.17 Performance Evaluation Using Confusion Matrix and Metrics

Fig 4.17 reports accuracy, precision, recall, F1-score, and mAP to quantitatively assess model effectiveness.

SNAPSHOTS (COCO)

4.18 Dataset Overview (COCO)

The model is trained and evaluated on the **COCO Dataset (2017 version)** using the full **train2017** and **val2017** splits without any subset selection.

- Training set: ~118,000 images
- Validation set: ~5,000 images
- Total classes: 80 object categories

The dataset contains diverse real-world scenes with multiple objects, varying scales, and complex backgrounds.



Fig 4.18 Sample images from COCO dataset

4.19 Before Preprocessing

Raw COCO images with annotations are visualized prior to preprocessing. The dataset exhibits varying object scales, complex backgrounds, and uneven class distribution across categories.

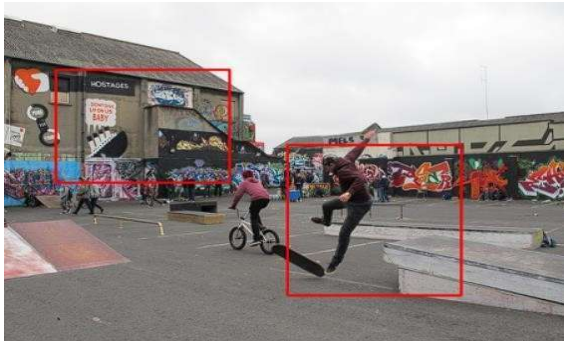


Fig 4.19 Raw images with bounding box annotations

4.20 After Preprocessing

Input images are resized to a fixed resolution (e.g., 512×512) and normalized for consistent model input. Annotations are reformatted to align with the resized dimensions, ensuring accurate bounding box representation.



Fig 4.20 Preprocessed input images

4.21 Attention Mechanism Visualization

Attention maps are visualized to highlight regions emphasized by the AttenRetinaNet model during detection. The model assigns higher focus to salient object areas while suppressing background noise, improving localization and classification performance.

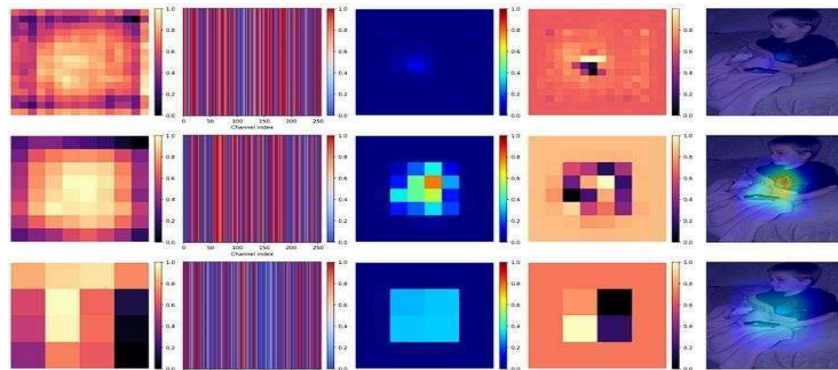


Fig 4.21 Attention maps highlighting object focus regions

4.22 Model Training Performance

The training performance is evaluated using the loss value across epochs. A consistent decrease in loss indicates effective learning and convergence of the AttenRetinaNet model.

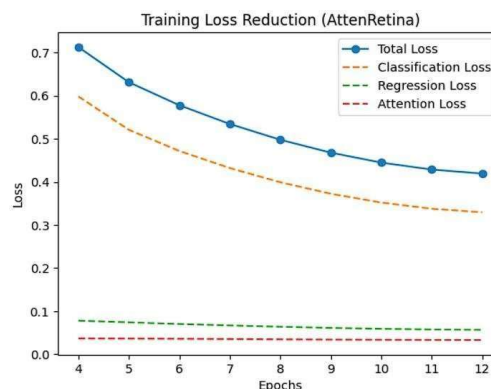


Fig 4.22 Training loss curve

4.23 Detection Results (Final Outputs)

The final detection results demonstrate the model's ability to accurately identify and localize multiple objects in complex scenes. The outputs include correctly detected objects, images with multiple instances, and challenging cases with varied scales and backgrounds.

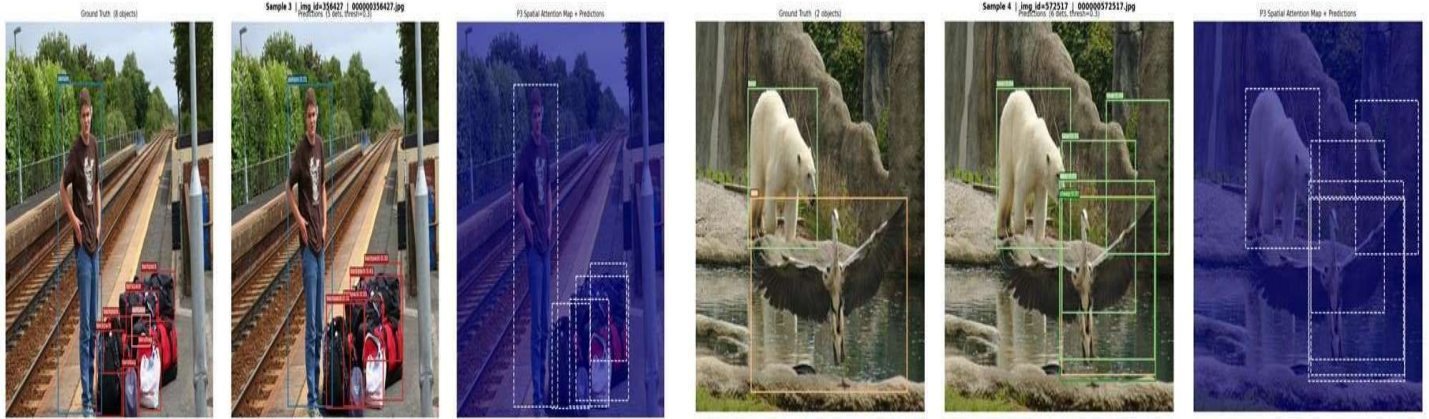
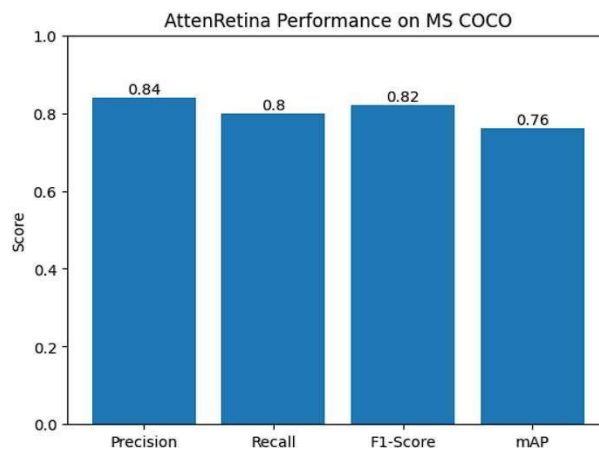


Fig 4.23 Final object detection results using AttenRetinaNet

4.24 Evaluation Metrics

The model performance is evaluated using standard COCO metrics, including mean Average Precision (mAP), precision, and recall. These metrics reflect the model's effectiveness in accurate object detection and localization.



Metric	Value
mAP	0.76
Precision	0.84
Recall	0.80

Table 1

Fig 4.24 Evaluation metrics in COCO dataset using AttenRetina

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The AttenRetina model provides an effective solution for object detection in autonomous driving systems. By integrating RetinaNet with Feature Pyramid Networks and attention mechanisms, the model significantly improves detection accuracy, especially for small and distant objects.

The system demonstrates strong performance in complex environments and achieves a good balance between accuracy and real-time processing. The use of multi-scale feature fusion and attention modules enhances the model's ability to detect objects under challenging conditions such as occlusion and varying lighting.

5.2 Future Work

- Optimize the model for real-time edge deployment.
- Reduce computational complexity.
- Improve detection in extreme weather conditions.
- Enhance attention mechanisms.
- Integrate with full autonomous driving systems

CHAPTER 6

REFERENCES

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CHAPTER 7

APPENDIX

Appendix A: Base Paper Summary

The base paper proposes the AttenRetina model, which combines RetinaNet with a Feature Pyramid Network and attention mechanisms. The model improves object detection accuracy by enhancing multi-scale feature representation and focusing on relevant regions within the image.

Appendix B: Dataset Details

The model is evaluated using standard datasets commonly used in object detection tasks.

- **KITTI Dataset:** Contains real-world driving scenarios with annotated objects such as cars, pedestrians, and cyclists. It is widely used for evaluating autonomous driving systems.
- **MS COCO Dataset:** A large-scale dataset containing diverse object categories with complex backgrounds. It is used to test the generalization ability of the model.
- The KITTI dataset consists of real-world driving scenes with annotations for vehicles, pedestrians, and cyclists. It contains approximately 7,481 training images and is widely used for evaluating autonomous driving systems.
- The MS COCO dataset contains over 118,000 training images with 80 object categories. It includes complex scenes with multiple objects, making it suitable for evaluating general object detection performance.
- These datasets help evaluate the model's ability to detect both small and large objects under varying environmental conditions.

Appendix C: Evaluation Metrics

The performance of the model is evaluated using standard object detection metrics:
Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7.1)$$

Precision measures how many of the predicted positive detections are actually correct. Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7.2)$$

Recall indicates how many actual objects are correctly detected by the model. F1-Score:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7.3)$$

F1-score provides a balance between precision and recall.
Mean Average Precision (mAP):

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (7.4)$$

where AP is the average precision for each class.mAP is the most important metric for object detection as it evaluates the model's overall performance across multiple classes.